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SUPERSEDING
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September 30, 1958

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK, BLEACHED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bleached cotton duck utilized in the manufacture of clothing items (see 6.1).

1.2 Classification.

1.2.1 *Types.* The bleached cotton duck shall be of the following types as specified (see 6.1 and 6.2).

Type I.—5 ounces per square yard.

Type II.—7 ounces per square yard (1 by 2 ply).

Type III.—8.2 ounces per square yard.

Type IV.—8.5 ounces per square yard.

Type V.—7.0 ounces per square yard (2 by 2 ply).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191—Textile Test Methods.

PPP-P-51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

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(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act effective March 3, 1960.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is required, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 *Yarn.* The cotton used in the manufacture of the yarn shall be thoroughly cleaned, carded, drawn, spun and twisted into singles or plied yarns as specified in table I.

3.3 Color. The color shall be bleached white. The use of fluorescent optical bleaches to complement the color of the bleach is permitted.

3.3.1 *Matching.* The color shall match the standard sample (see 6.3) under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

3.5 Weave. The weave for types I, III and V cloths shall be plain, one up, one down. The weave for types II and IV cloths shall be an oxford weave with two ends weaving as one and alternating in plain-weave order with each pick.

3.6 Finish. The cloth shall be desized, singed, bleached, and mercerized. Total re-

TABLE I.—*Physical requirements*

Type	Weight per square yard minimum	Yarn ply		Yarns per inch minimum		Breaking strength minimum	
		Warp	Filling	Warp	Filling	Warp	Filling
	Ounces					*Pounds*	*Pounds*
I	5.0	2	2	77	44	130	75
II	7.0	1	2	88	28	110	60
III	8.2	2	2	58	48	135	100
IV	8.5	1	4	112	32	120	120
V	7.0	2	2	52	34	80	48

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sidual size, finishing or nonfibrous material, shall not exceed 2 percent when tested as specified in 4.4.

3.7 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 2 percent in either the direction of the warp or filling when tested as specified in 4.4. The preshrinking process used shall not be identified in any manner on the cloth, the ticket or on the package.

3.8 Width. The width shall be as specified (see 6.2) and shall be the minimum acceptable width exclusive of selvages.

3.9 Length. Unless otherwise specified (see 6.2), the cloth shall be put up on rolls in accordance with Federal Specification PPP–P–51. The cloth shall be in one continuous roll of not less than 50 yards.

3.10 Fiber identification. Each piece or roll of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act effective March 3, 1960.

3.11 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed that set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination for acceptance. Except when otherwise specified by civil government agencies, sampling and inspection shall be in accordance with the provisions of Military Standard MIL–STD–105, except where otherwise indicated. The cloth shall be examined to determine conformance with the requirements of this specification.

4.2.1 *Examination of the end item.* Defects found during this examination shall be classified in accordance with 4.2.1.1, 4.2.1.2 and 4.2.1.3.

4.2.1.1 *Yard-by-yard examination.* The defects listed in table II shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality levels (AQL's) for this examination shall be 2.5 major and 10.0 total (major and minor combined) defects per 100 units (yards). The sample size shall be based upon inspection level II of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table III. An approximately equal number of yards shall be examined in each piece of the sample.

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TABLE II.—*Classification of defects*

Glossary Numbers Fed. Std. No. 4	Defect	Classification	
		Major	Minor
90, 91, 92	Hole, cut, tear or break in body of cloth	X	
78	Smash—any	X	
11, 33	Broken or missing yarns:		
	Three or more contiguous regardless of length	X	
	Two contiguous, 2 inches or more missing	X	
	Two contiguous, less than 2 inches or single, 2 inches or more missing		X
6, 28	Kinks—single, more than 1 inch in length or multiple 1 inch or less in length		X
7, 29	Knot, oversize, or untrimmed ends extending from surface		X
40, 41	Wrong draw, reed mark—clearly visible[1]		X
42	Temple mark—resulting in clearly visible marks within minimum specified width of cloth		X
49, 50	Selvage tight—resulting in pronounced slackness or waviness in cloth		X
55, 56	Selvage broken, cut or torn	X	
51	Selvage scalloped—indentation exceeding ¼ inch		X
93	Baggy, ridgy, or wavy cloth clearly visible[1]	X	
89	Tender or weak place—any	X	
79	Abrasion mark—clearly visible,[1] fuzziness but not resulting in a weak spot		X
69, 70, 74, 75, 76[2]	Spot, stain or streak:		
	Clearly visible[1] and 3 inches or more in combined directions	X	
	Clearly visible[1] and less than 3 inches in combined directions		X
14, 57, 59, 61	Mixed filling, finish mark, embedded wrinkle or crease—clearly visible[1]	X	
36	Mixed end		X

[1] Clearly visible at normal inspection distance (approximately 3 feet).

[2] Except defects of spot, stains or streak shall be counted regardless of cause, i.e., not restricted to rust, oil or grease.

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TABLE II.—*Classification of defects (cont'd)*

Glossary Numbers Fed. Std. No. 4	Defect	Classification	
		Major	Minor
9, 12, 35	Fine filling bar or light place—clearly visible[1]	X	
	Open place, crack, or hitchback—opening at widest point three or more filling yarn diameters or sequence of openings resulting in a weak spot	X	
	Clearly visible,[1] but opening at widest point less than three filling yarn diameters, or raised places resulting from slack warp ends		X
13, 34	Floats or skips: Multiple, ½ inch or more in combined warp and filling directions, or harness skips	X	
	Multiple, less than ½ inch in combined warp and filling directions or single floating over 2 inches or more		X
1, 2, 4, 10, 26	Heavy filling bar, heavy place, coarse or additional yarns, clearly visible[1]		X
3, 5, 27, 77	Slough-off, jerked-in-filling, slugs, slubs, gouts in either direction, three or more times the thickness of normal filling yarns	X	
	Clearly visible[1] but not more than three times the thickness of the normal filling yarn		X

[1] Clearly visible at normal inspection distance (approximately 3 feet).

4.2.1.2 *Over-all examination.* Each defect listed below shall be counted not more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the acceptance number (number of defects acceptable) shall be as shown in table III.

Defects

(*a*) Selvage curled, folded, rolled, or slack, continuously or intermittently throughout the piece.
(*b*) Selvage beaded or corded continuously or intermittently throughout the piece.
(*c*) Over-all uncleanness, objectionable odor.
(*d*) Width less than specified.
(*e*) Uneven weaving throughout piece, clearly visible at normal inspection distance of 3 feet.
(*f*) Preshrinking process identified on cloth.
(*g*) Fiber identification not shown on piece.

TABLE III.—*Sample size*

Lot size in yards	Sample size in pieces	Acceptance number
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 to 550,000	25	1
550,001 and up	50	2

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.1.3 *Length examination.*

4.2.1.3.1 *Examination for length of individual pieces.* Each piece in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum length or more than the maximum

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gross length permitted, or any gross length found to be more than 2 yards less than the gross length marked on the ticket shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The sample size and acceptance number shall be as shown in table III.

4.2.1.3.2 *Examination for total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of the pieces in the sample is less than the total gross lengths marked on the piece tickets. The pieces examined shall be those selected for examination of individual pieces.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping roll fully prepared for delivery. The lot size shall be the number of shipping rolls in the inspection lot. The inspection level shall be L–4 and the AQL shall be 4.0 defects per hundred units.

Examination	Defect
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application. Preshrinking process identified on roll.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of rolls, inadequate securing or sealing of wrapping materials, insecure sewing or loose wire ties.
Weight	Gross weight beyond the limits specified.

4.4 Testing of the end item. The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 3 continuous yards full width of the finished fabric. The sample size for each characteristic shall be as listed in table IV:

TABLE IV.—*Sample size*

Lot size in yards	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

The lot shall be unacceptable if one or more units tested fail to meet any of the specified test characteristics. When the data in the "number of determinations per sample unit" and the "results reported as" columns are not specified in table V they shall be as required by the referenced test method.

5. PREPARATION FOR DELIVERY

(Federal Standard No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civilian procurements.)

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cloth put up on rolls, shall be packaged in accordance with the applicable requirements for class 4 fabrics of Federal Specification PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51.

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TABLE V.—*Test methods*

Characteristics	Requirement reference	Test method	Number of determinations per individual unit of product	Results reported as
Weight, ounce/square yard	Table I	5041		
Yarns per inch	Table I	5050		
Breaking strength	Table I	5100		
Mercerization	3.6	Microscopic	1	Pass or fail
Nonfibrous material[1]	3.6	2611		
Shrinkage in laundering[2]	3.7	5550	3	Average of 3 specimens to nearest 0.1 percent
Weave	3.5	Visual	1	Pass or fail
Yarn ply	Table I	Visual	1	Pass or fail

[1] The total nonfibrous material shall be determined by method 2611 utilizing the following formula:

$$\text{Nonfibrous materials content, percent} = \frac{O-S}{O} \times 100$$

[2] The three specimens required by shrinkage test shall be selected from the unit of product as follows: One specimen from each side of the cloth within three inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns. The shrinkage of the specimen shall be calculated as follows:

$$\text{Shrinkage, percent} = \frac{A-B}{A} \times 100$$

Where A = Average of initial measurements (3 specimens).
B = Average of measurements after laundering (3 specimens).

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use. Types I, II, and III bleached cotton ducks covered by this specification are intended for use in white uniforms for personnel of the Department of Defense. Type IV bleached cotton duck is intended for use in dress uniform trousers of U.S. Military Academy Cadets. Type V bleached cotton duck is utilized as the base fabric for Smock, Man's, Dental Operating, and for Smock, Man's, Medical Assistant.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.
(*b*) Type and nomenclature of cloth required (see 1.2).
(*c*) Width required (see 3.8).
(*d*) Length of pieces if other than specified (see 3.9).
(*e*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives as indicated in Federal Specification PPP–P–51 may be exercised in the preparation for delivery of cloth.

6.3 Standard sample. For access to standard sample address the procuring office issuing the invitation for bids.

6.4 Transportation description. The transportation descriptions and minimum weights applicable to this item are:

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Rail:

Cotton piece goods in the original piece
Carload minimum weight, any quantity.

Motor:

Cotton piece goods not otherwise indexed
Motor volume minimum weight, any quantity.

6.5 Level B packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Copies of this specification may be purchased

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